

Exploratory Data Analysis (EDA) on Titanic Dataset

1. Executive Summary:

This report presents exploratory data analysis (EDA) on the Titanic dataset, aiming to uncover patterns and relationships that may influence passenger survival. The analysis focuses on understanding the distribution of key features such as age, fare, class, and sex, as well as visualizing correlations between numerical variables.

2. Introduction:

The Titanic dataset is a well-known dataset used for classification problems. It contains data on passengers such as age, sex, class, fare paid, and whether they survived or not. This analysis helps identify important factors that may contribute to survival rates, preparing the data for future modelling or decision-making.

3. Tools and Libraries:

This project was carried out using:

- **Python**
- **Pandas** – for data manipulation
- **Seaborn & Matplotlib** – for visualization
- **NumPy** – for numerical operations

4. Data Overview:

- The dataset was loaded using pandas.
- Initial inspections included:
 - `.info()` to check data types and null values.
 - `.describe()` to understand statistical distributions.
 - `.nunique()` and `.value_counts()` to explore categorical data.

Key Insights:

- Age: 177 missing values
- Cabin: 687 missing values (majority)
- Embarked: 2 missing values
- Sex and Embarked show categorical imbalance

5. Visual Exploration:

◆ Pairplot

A pairplot was used to visualize pairwise relationships between numeric variables like Age, Fare, SibSp, and Parch.

◆ Correlation Heatmap

A heatmap revealed moderate correlation between:

- Fare and Pclass (negative)
- Parch and SibSp (positive)

These insights help understand how strongly variables relate.

6. Focused Visuals:

Age vs Fare

A scatterplot was used to identify any trends between ticket price and passenger age. This can help infer if younger or older passengers tended to afford higher-class tickets.

Sex and Class Distribution

Bar plots were created to visualize:

- Count of male vs. female passengers
- Count of passengers per travel class (1st, 2nd, 3rd)

These visuals show strong gender and class-based patterns in the dataset.

7. Conclusion:

The Titanic dataset shows clear differences in passenger characteristics across gender, class, and fare. Missing values in key columns like Age and Cabin will need handling for any future predictive modelling. This EDA provides a solid foundation for feature selection and machine learning preparation.